

Proposal for AI-Powered Online Quran Teaching & Quality Assurance Platform



Executive Summary

This proposal outlines the development of a specialized Online Quran Teaching & Quality Assurance Platform designed for Quran academies, Islamic educational institutions, and online Quran tutoring organizations.

The platform will enable academies to conduct live Quran classes, automatically record sessions, generate lesson transcripts, evaluate teaching quality through Artificial Intelligence, monitor curriculum adherence, and provide detailed reports to academy management.

The objective is to improve educational standards, ensure consistency in teaching methodologies, enhance student learning outcomes, and provide academy administrators with meaningful insights into class performance and instructional quality.

Project Vision

To establish a modern, technology-driven Quran learning environment that combines online education with intelligent quality assurance, ensuring every student receives high-quality Islamic education while maintaining the academy's teaching standards.

Key Objectives

- Conduct secure online Quran classes.
- Record all classes automatically.
- Generate lesson transcripts after each class.
- Evaluate lesson quality using AI.
- Ensure teachers follow the assigned curriculum.
- Monitor lesson effectiveness and student engagement.
- Generate automated reports for academy administrators.
- Maintain a complete archive of recorded classes.
- Support continuous improvement of teaching standards.

Proposed Solution

The platform will provide an end-to-end workflow for Quran education management.

Class Workflow

1. Academy schedules Quran classes.
2. Teachers conduct live sessions.
3. Sessions are automatically recorded.
4. Recordings are uploaded to secure cloud storage.
5. Audio is converted into text.
6. AI analyzes lesson content.
7. A quality assurance report is generated.
8. Administrators review reports and recordings.

Core Features

Administration Portal

- Teacher Management
- Student Management
- Course Management
- Quran Curriculum Management
- Class Scheduling
- Attendance Monitoring
- AI Quality Assurance Reports
- Recording Archive
- Teacher Performance Analytics
- Student Progress Insights

Teacher Portal

- Daily Teaching Schedule
- Live Class Management
- Student Attendance
- Lesson History
- Teaching Performance Reports
- Recording Access

Student Portal

- Join Quran Classes
- Attendance History

- Access Recorded Lessons
 - Learning Progress Overview
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AI-Powered Lesson Evaluation

Following each class, Artificial Intelligence will analyze the lesson transcript and generate an educational quality report.

Curriculum Adherence

The system will evaluate:

- Whether the assigned lesson was covered
- Topic completion status
- Lesson continuity
- Curriculum progression

Teaching Quality Assessment

The system will assess:

- Explanation clarity
- Teaching structure
- Lesson organization
- Time utilization
- Educational effectiveness

Quran Lesson Analysis

Depending on the course type:

Nazira Quran

- Lesson coverage verification
- Recitation practice duration
- Student participation indicators

Tajweed

- Coverage of Tajweed topics
- Rule explanation assessment
- Practical examples discussed

Hifz-ul-Quran

- Revision activities

- Memorization session duration
- Daily lesson coverage

Islamic Studies

- Topic relevance
- Educational content quality
- Discussion structure

Teacher Performance Reports

Each class will generate:

Class Information

- Teacher Name
- Student Name(s)
- Date and Time
- Lesson Duration

Attendance Summary

- Teacher Attendance
- Student Attendance
- Participation Metrics

Lesson Evaluation

- Curriculum Adherence Score
- Teaching Effectiveness Score
- Lesson Completion Score
- Engagement Score

AI Summary

- Topics Covered
- Key Teaching Points
- Lesson Highlights
- Improvement Recommendations

Recording Access

- Recording Link
- Transcript Access
- Report Download

Student Progress Support

The platform can provide insights into:

- Attendance consistency
 - Lesson completion history
 - Revision frequency
 - Learning continuity
 - Class participation patterns
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Recording & Archive Management

Every class session will:

- Be recorded automatically
 - Be stored securely
 - Be uploaded to Google Drive
 - Be associated with the relevant teacher and student
 - Be accessible for future review
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Benefits for the Academy

Academic Benefits

- Standardized teaching quality
- Improved lesson consistency
- Better curriculum tracking
- Enhanced educational outcomes

Administrative Benefits

- Reduced manual supervision
- Automated lesson evaluation
- Centralized reporting
- Historical class archive

Teacher Development

- Constructive performance feedback
- Continuous professional improvement
- Teaching effectiveness insights

Student Experience

- Higher lesson quality
 - Improved learning consistency
 - Access to lesson recordings
 - Better educational support
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Technology Platform

Frontend

- Next.js 15
- TypeScript
- Tailwind CSS

Backend

- NestJS
- PostgreSQL
- Prisma ORM

Online Classes

- LiveKit Video Platform

AI Services

- OpenAI / Gemini
- Whisper Speech-to-Text / Google TTS

Storage

- Google Drive Integration

Infrastructure

- Docker
 - Redis
 - BullMQ
 - Nginx
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Security & Privacy

The platform will implement:

- Role-Based Access Control
- Secure Authentication
- Encrypted Data Transmission
- Secure Recording Storage
- Audit Logs
- Controlled Recording Access
- Data Backup & Recovery

Project Deliverables

1. Admin Dashboard
2. Teacher Dashboard
3. Student Dashboard
4. Live Quran Classroom System
5. Attendance Management
6. Recording Management
7. Google Drive Integration
8. Speech-to-Text Processing
9. AI Lesson Evaluation Engine
10. Teacher Performance Reports
11. Student Progress Reports
12. PDF Report Export
13. Deployment & Documentation

Estimated Timeline

Phase	Description	Duration
Phase 1	Planning & Architecture	5 Days
Phase 2	User Management & Authentication	5 Days
Phase 3	Class Scheduling & Curriculum Management	10 Days
Phase 4	Live Classroom Integration	3 Days
Phase 5	Recording & Cloud Storage	3 Days
Phase 6	Transcript Generation	2 Days
Phase 7	AI Evaluation Engine	5 Days
Phase 8	Reporting & Analytics	1 Day
Phase 9	Testing & Quality Assurance	1 Week
Phase 10	Deployment & Training	1 Week

Estimated Project Duration: 5–7 Weeks

Budget Date frame

Phase	Description	Due	Cost
Phase 1	Initial Development & VPS setup	Project Starting Date	\$200
Phase 2	Integrating Online Classes with AI & Google storage for persistence data	Under Development (2 weeks)	\$200
Phase 3	Project Finalized & Deployed to Production.	Finalizing & Deployment	\$200

Estimated Project Budget: \$600 (Approx. PKR. 167,130 Only)

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